

Analytical CMP Simulator

Physics, Kinematics, Tribology & Mathematical Models

A comprehensive deep-dive breakdown of the physics, kinematics, tribology, and mathematical models governing the Chemical Mechanical Polishing (CMP) Simulator.

The engine operates by numerically integrating the classic Preston Equation over a dynamic vector field generated by an oscillating four-bar linkage mechanism.

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1 Kinematic Modeling (The Four-Bar Linkage)

The foundation of the simulator is the motion of the polishing arm. The machine uses a crank-rocker mechanism to translate continuous motorized rotation into an oscillating sweep across the base polishing pad.

1.1 Vector Loop & Position Geometry

The system is modeled on a Cartesian coordinate plane where all physical components are mapped dynamically.

- **Crank Pivot:** Fixed at the origin $(0, 0)$.
- **Rocker/Hub Pivot (D):** Fixed at coordinates $(D_x, D_y) = (360, 0)$ mm.
- **Crank Angle (θ_2):** Driven continuously by the input crank speed ω_c (Crank RPM).

The instantaneous position of the crank-coupler joint B is defined as:

$$B_x = L_2 \cos(\theta_2) \quad (1)$$

$$B_y = L_2 \sin(\theta_2) \quad (2)$$

The distance d between the moving joint B and the fixed rocker pivot D changes continuously:

$$d = \sqrt{(D_x - B_x)^2 + (D_y - B_y)^2} \quad (3)$$

To find the position of the coupler-rocker joint C , the engine calculates the geometric intersection of two circles: one centered at B with a radius equal to the coupler link L_3 , and another centered at D with a radius equal to the rocker link L_4 . Using the Law of Cosines, the intersection height h and projected base distance a along line BD are:

$$a = \frac{L_3^2 - L_4^2 + d^2}{2d} \quad (4)$$

$$h = \sqrt{\max(0, L_3^2 - a^2)} \quad (5)$$

Reasoning for Constraints: The $\max(0, \dots)$ function acts as a critical programmatic safeguard against imaginary numbers. If $L_3^2 - a^2 < 0$, it means the links cannot physically span the distance required to connect. This indicates a **Kinematic Lockup** state, causing the simulator to flag a mechanical lockup error and halt the physics engine.

1.2 Tool Center & Arm Projection

Once joint C is solved, the absolute rocker angle θ_4 (the angle of the rocker arm hub) is determined. The actual polishing tool is mounted at the end of a long, rigid arm extending directly from this oscillating hub:

$$Arm_{angle} = \theta_4 - \alpha_{off} + (\pi \text{ if flipped}) \quad (6)$$

$$Tool_x = D_x + A_{len} \cos(Arm_{angle}) \quad (7)$$

$$Tool_y = D_y + A_{len} \sin(Arm_{angle}) \quad (8)$$

1.3 Eccentricity & Sweep Velocity

The most critical spatial metric in CMP modeling is the instantaneous eccentricity $e(t)$, which tracks the absolute radial distance from the center of the rotating base pad (360, 520) to the center of the top polishing tool:

$$e(t) = \sqrt{(Tool_x - 360)^2 + (Tool_y - 520)^2} \quad (9)$$

Because the analytical derivative of this complex four-bar projection is highly non-linear, the engine evaluates the instantaneous radial sweep velocity e_{dot} using a central finite-difference method over a highly discrete timestep Δt :

$$e_{dot} = \frac{e(t + \Delta t) - e(t - \Delta t)}{2\Delta t} \quad (10)$$

Reasoning for Calculation: The value of e_{dot} is critical because as the arm sweeps across the pad, its radial velocity adds a linear velocity component directly onto the rotational velocity vectors, directly skewing the uniform material removal profile.

2 Contact Mechanics & Friction Drive Physics

2.1 Area of Overhang ($Area_{eff}$)

When the four-bar kinematics force the tool to swing near the outer boundaries of the base pad, the tool may partially overhang the edge. This reduces the true contact area and dramatically alters pressure distributions.

If the condition $e + r_t > R_b$ is met, the tool is overhanging, and the effective area ratio $area_{eff}$ is computed by calculating the intersection area of two asymmetrical circles:

$$A_{intersect} = r_t^2 \arccos\left(\frac{e^2 + r_t^2 - R_b^2}{2e r_t}\right) + R_b^2 \arccos\left(\frac{e^2 + R_b^2 - r_t^2}{2e R_b}\right) - \frac{1}{2}\sqrt{s_{val}} \quad (11)$$

Where s_{val} is derived from Heron's formula for the triangle formed by the two centers and their points of intersection. The final ratio is clamped safely:

$$area_{eff} = \max\left(0.01, \frac{A_{intersect}}{\pi r_t^2}\right) \quad (12)$$

2.2 Localized Contact Pressure (P)

Standard manufacturing models assume uniform pressure across the wafer interface. However, industrial polishing pads are heavily grooved, meaning only a fraction of the nominal area makes physical contact. The simulator accounts for this via the pad contact parameter:

$$P_0 = \frac{(Mass_{tool} + F_{down}) \cdot g}{\pi r_t^2} \quad (13)$$

$$P_{inst} = \frac{P_0}{area_{eff} \cdot \left(\frac{padContactPct}{100} \right)} \quad (14)$$

Reasoning for Calculation: If a pad is only **60%** ungrooved (flat), the entire downforce is compressed into a smaller surface area, elevating the real localized contact pressure linearly and accelerating the material removal rate.

2.3 Freewheel / Friction Drive Torque Balance

When the top tool is set to **Freewheel** mode, it has no active motor. Instead, it is rotationally driven entirely by viscous slurry shearing and direct friction from the base pad spinning beneath it. The engine models this glass block dynamically ($\rho = \mathbf{2500 \text{ kg/m}^3}$, **25 mm** thick).

The induced top tool RPM (N_t) is governed by the base pad RPM (N_p) minus an operational drag-induced slip velocity:

- **Coupling Stiffness:** Proportional to $area_{eff} \times \mu$ (Slurry Friction Coefficient).
- **Drag Torque Factor:** Increases as a function of the total downforce mass and cross-sweep velocity ($1.0 + 2.0 |e_{dot}|$).

$$N_{t(ideal)} = N_p \cdot (1.0 - (1.0 - area_{eff}) \cdot 0.20) \quad (15)$$

$$N_t = N_{t(ideal)} - \left(\frac{Drag}{Stiffness} \times C_{scale} \right) \quad (16)$$

Reasoning for Calculation: When the tool sweeps quickly (e_{dot} is high) or hangs over the edge ($area_{eff}$ drops), the friction coupling degrades. The top disc “slips” or lags behind the base pad’s rotation speed, completely warping the relative velocity field.

3 Material Removal Physics (The Preston Model)

The core tribological behavior of the simulator is driven entirely by the classic **Preston Equation**:

$$MRR = K_p \cdot P \cdot V_{rel} \quad (17)$$

To compute the precise amount of material removed across the entire surface profile, the engine must numerically integrate the relative velocity vector field V_{rel} of every micro-point across a full 360° closed-loop cycle of the crank.

3.1 Top Disc Kinematics (Wafer / Tool Wear)

For any given radius r_m on the *top tool*, the engine evaluates the velocity vectors across multiple angular coordinates θ around that radius. The localized relative velocity components against the

pad are:

$$v_x = e_{dot} - \Omega \cdot r_m \sin(\theta) \quad (18)$$

$$v_y = \omega_p \cdot e + \Omega \cdot r_m \cos(\theta) \quad (19)$$

Where $\Omega = \omega_t - \omega_p$ represents the differential rotational velocity between the two platforms. The absolute relative velocity is:

$$V_{rel} = \sqrt{v_x^2 + v_y^2} \quad (20)$$

This vector field is integrated across 32 angular steps $d\theta$, and then calculated over 360 individual crank positions to yield the total historical material removal depth at that specific tool radius.

3.2 Base Disc Kinematics (Polishing Pad Wear)

Computing wear on the *base pad* requires an inverse reference frame transformation. For a selected radius R_b on the base pad, a specific point only undergoes wear when the moving tool happens to sweep directly over it.

The engine calculates the instantaneous intersection chord angle ϕ of the base pad annulus hidden beneath the tool:

$$\phi = \arccos\left(\frac{R_b^2 + e^2 - r_t^2}{2 R_b e}\right) \quad (21)$$

The relative velocity vectors are calculated strictly for the overlapping arc window (spanning from $-\phi$ to $+\phi$), averaged, and then scaled by the total contact duty cycle $\left(\frac{\phi}{\pi}\right)$ across the pad's path.

4 Surface Topography & Conic Curve Fitting

Multiplying the calculated Material Removal Rate (MRR) by the operational polishing time yields the **Depth of Cut (DOC)** in micrometers. To mathematically characterize the resulting geometric lens profile, the simulator performs a high-resolution Least Squares Linear Regression against a second-degree polynomial (conic section equation):

$$Z(r) = ar^2 + c \quad (22)$$

By substituting variables such that $x = r^2$ and $y = Z$:

$$a = \frac{n \sum(xy) - \sum x \sum y}{n \sum x^2 - (\sum x)^2} \quad (23)$$

4.1 Topography Form Reasoning

- **Concave vs. Convex Forms:** If the calculated coefficient $a < 0$, the center of the disc is eroding faster than its outer periphery, giving rise to a hollowed **Concave** profile. If $a > 0$, the outer edges roll off faster, shaping a crowned **Convex** profile.

- **Conic Eccentricity Classification** (e_{conic}): The curvature is categorized according to standard analytic geometry:
 - Deviation $\approx 0 \rightarrow e_{conic} = 0$ (**Perfect Circle / Flat Surface**)
 - Minor Deviation $\rightarrow 0 < e_{conic} < 1$ (**Elliptic Profile**)
 - Moderate Deviation $\rightarrow e_{conic} = 1$ (**Parabolic Profile**)
 - Heavy Deviation $\rightarrow e_{conic} > 1$ (**Hyperbolic Profile**)

5 AI Intelligent Tuning Optimization Architecture

The automated optimization engine calculates the exact mechanical linkage lengths and rotational speeds required to correct a warped or unflat disk profile. This operates via variance-covariance correlation tracking.

Given an initial measured base profile dataset $H_{initial}$ and an analytically simulated wear vector W , the optimizer isolates the exact polishing time t required to perfectly match and level out the high peaks down to the lowest valley floor:

$$t = \frac{\sum(H_{initial} - \bar{H}) \cdot (W - \bar{W})}{\sum(W - \bar{W})^2} \quad (24)$$

The deep search routine rapidly generates thousands of distinct kinematic candidate machines. For every candidate evaluated:

1. It maps out the theoretical base pad wear profile W .
2. It projects the final corrected topography: $H_{final} = H_{initial} - (W \cdot t)$.
3. It extracts the raw Peak-to-Valley (PV) error of H_{final} .
4. **Top Disc Flatness Guard Constraint:** It simultaneously checks the wear profile generated on the *top tool*. If a specific configuration flattens the base pad but severely crowns or cups the top wafer, it flags a violation and applies a massive penalty score:

$$Penalty = Error_{top_flatness} \times PV_{initial} \times 10.0 \quad (25)$$

The candidate configuration that yields the lowest combined final PV error and top disc wear penalty is chosen and presented as the ideal plan.

6 Nomenclature Table

Symbol	Description	Standard Units / Scale
L_2	Crank link length (driving radius)	mm
L_3	Coupler link length	mm
L_4	Rocker link length	mm
D_x, D_y	Ground coordinates of fixed rocker axis pivot	mm
A_{len}	Extension length of the rigid polishing arm	mm

Symbol	Description	Standard Units / Scale
α_{off}	Hub Angle (angular mounting offset of arm from rocker)	degrees
θ_2	Instantaneous angular position of the crank	radians / degrees
θ_4	Instantaneous angular position of the rocker hub	radians / degrees
$e(t)$	Instantaneous Eccentricity (center-to-center distance)	mm
e_{dot}	Radial sweep velocity of the tool center	m/s
ω_c	Angular velocity of the driving crank link	rad/s
ω_p, N_p	Angular velocity / RPM of the base pad	rad/s, RPM
ω_t, N_t	Angular velocity / RPM of the top polishing tool	rad/s, RPM
Ω	Relative angular velocity differential ($\omega_t - \omega_p$)	rad/s
r_t	Physical radius of the top polishing tool	m, mm
R_b	Physical radius of the main base pad	m, mm
F_{down}	Superimposed active downward pneumatic force	kg
$area_{eff}$	Effective percentage area of tool remaining on pad	Dimensionless ratio (0.01–1.0)
P_0	Nominal static downward contact pressure	Pa (N/m^2)
P_{inst}	Micro-scale instantaneous real contact pressure	Pa (N/m^2)
K_p	Preston constant (slurry/material coefficient)	$m^2/N \cdot 10^{-13}$
V_{rel}	Magnified localized relative velocity vector	m/s
MRR	Material Removal Rate depth velocity	$\mu m/hr$
$Z(r)$	Radial surface topography depth profile function	μm
a	Second-order conic section curvature coefficient	Dimensionless
e_{conic}	Conic section geometric eccentricity value	Dimensionless